

Heckscher-Ohlin Model

ECON 404: Cal Poly

Motivation

- Model developed by Eli Heckscher and his student Bertil Ohlin in the early 1900s.
- At the time, there was massive amounts of trade – including machinery.
 - This ability to transport machines meant harder to justify differences in technology was the driving force of trade (like Ricardo)
- So, they assume there are differences in resources.
- Even today:
 - Canada – lots of land
 - U.S, Western Europe, and Japan – lots of skilled labor
 - China and other Asian countries – lots of unskilled labor

Model Setup

- Consider two sectors; Computers (C) and Shoes (S).
- Capital (K) and Labor (L) are used in both sectors with the total amount of Capital and Labor fixed.
- Home:

$$\bar{K} = K_C + K_S$$

$$\bar{L} = L_C + L_S$$

- Foreign:

$$\bar{K}^* = K_C^* + K_S^*$$

$$\bar{L}^* = L_C^* + L_S^*$$

Assumptions

- 1 Both factors can move freely between industries.
- 2 Shoe production is labor-intensive; that is, it requires more labor per unit of capital to produce shoes than computers

$$\frac{L_S}{K_S} > \frac{L_C}{K_C}$$

- 3 Foreign is labor-abundant. Equivalently, Home is capital abundant

$$\frac{\bar{L}^*}{\bar{K}^*} > \frac{\bar{L}}{\bar{K}} \quad \text{or} \quad \frac{\bar{K}}{\bar{L}} > \frac{\bar{K}^*}{\bar{L}^*}$$

Assumptions

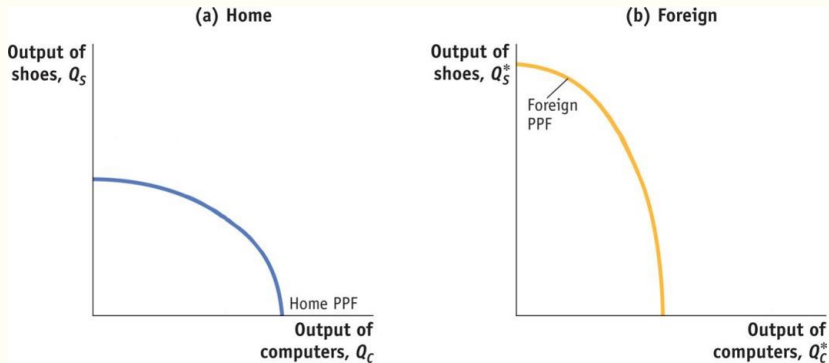
- 4 The final outputs can be traded freely, but labor and capital do not move between countries.
- 5 The technologies used to produce the two goods are identical across the countries – i.e. everyone produces the same way.

$$Q_C = Q_C(K_C, L_C)$$

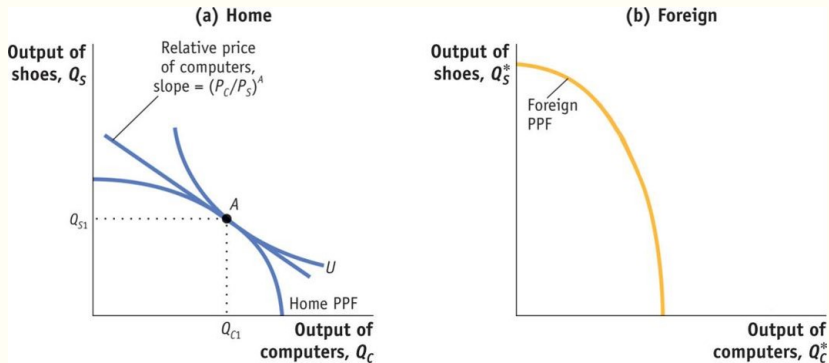
$$Q_S = Q_S(K_S, L_S)$$

- 6 Consumer tastes are the same across countries, and preferences for computers and shoes do not vary with the country's level of income.

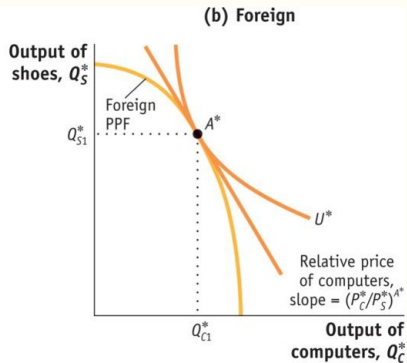
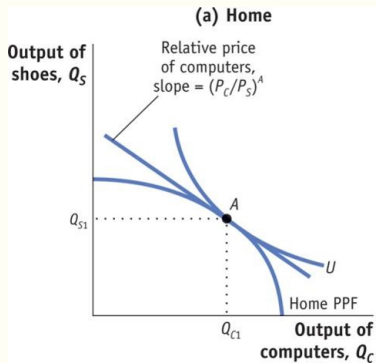
No Trade Equilibrium



No Trade Equilibrium



No Trade Equilibrium



Choosing the Mix of Inputs – Home

- Again, with perfect competition, we know that in equilibrium

$$P_C MPL_C = w = P_S MPL_S \quad (1)$$

$$P_C MPK_C = r = P_S MPK_S \quad (2)$$

From these equations we can see that

$$\begin{aligned} \frac{P_C}{P_S} &= \frac{MPL_S}{MPL_C} = \frac{MPK_S}{MPK_C} \\ \Rightarrow \frac{MPK_C}{MPL_C} &= \frac{MPK_S}{MPL_S} \end{aligned}$$

- Thus, the slope of the PPF is either $-\frac{MPL_S}{MPL_C}$ or $-\frac{MPK_S}{MPK_C}$.

Finding Input Intensity

- **Labor (capital) intensive:** A good is labor (capital) intensive relative to another good if its production requires more (less) labor per machine than the other good requires in its production.
- **Example:** Suppose production is Cobb-Douglas

$$Q_S = L_S^{\frac{3}{4}} K_S^{\frac{1}{4}} \quad (3)$$

$$Q_C = L_C^{\frac{1}{4}} K_C^{\frac{3}{4}} \quad (4)$$

Find the marginal products and technical rate of substitution (TRS)

$$\frac{MPK_S}{MPL_S} = \frac{L_S}{3K_S}$$

Shoes

$$\frac{MPK_C}{MPL_C} = \frac{3L_C}{K_C}$$

Computers

Finding Input Intensity

- In Equilibrium the TRS in each industry has to be equal.
Why?

$$\begin{aligned}\frac{MPK_C}{MPL_C} &= \frac{MPK_S}{MPL_S} \\ \frac{L_S}{3K_S} &= \frac{3L_C}{K_C} \\ \Rightarrow \frac{L_S}{K_S} &= 9 \left(\frac{L_C}{K_C} \right)\end{aligned}$$

- Therefore, the Shoe sector will require 9 times as much labor pre capital as the Computer sector making Shoes *labor intensive* and Computers *capital intensive*.

Choosing the Mix of Inputs – Home

- Note that the country is constrained by the total amount of labor and capital. In the previous section we figured out the optimal mix of labor and capital, but not the exact amount used.
- We need our input constraints, but also Prices.
- Using our optimality conditions

$$\begin{aligned}\frac{P_C}{P_S} &= \frac{MPL_S}{MPL_C} = 3 \left(\frac{K_S}{L_F} \right)^{\frac{1}{4}} \left(\frac{L_C}{K_C} \right)^{\frac{3}{4}} \\ &= 3 \left(\frac{K_C}{9L_C} \right)^{\frac{1}{4}} \left(\frac{L_C}{K_C} \right)^{\frac{3}{4}} \\ &= \left(\frac{3L_C}{K_C} \right)^{\frac{1}{2}}\end{aligned}$$

Choosing the Mix of Inputs – Home

- In order to pin down each amount of inputs used in each sector, we have four equations and four unknowns.

$$\frac{L_S}{K_S} = 9 \left(\frac{L_C}{K_C} \right) \quad (5)$$

$$\frac{P_C}{P_F} = \left(\frac{3L_C}{K_C} \right)^{\frac{1}{2}} \quad (6)$$

$$\bar{K} = K_C + K_S \quad (7)$$

$$\bar{L} = L_C + L_S \quad (8)$$

Choosing the Mix of Inputs – Home

- Solving the above system of equations yields:

$$\begin{aligned}K_S &= \frac{1}{8} \left(\frac{3\bar{L}}{\left(\frac{P_C}{P_S}\right)^2} - \bar{K} \right) & K_C &= \frac{3}{8} \left(3\bar{K} - \frac{\bar{L}}{\left(\frac{P_C}{P_S}\right)^2} \right) \\L_S &= \frac{9}{8} \left(\bar{L} - \frac{\bar{K} \left(\frac{P_C}{P_S}\right)^2}{3} \right) & L_C &= \frac{1}{8} \left(3\bar{K} \left(\frac{P_C}{P_S}\right)^2 - \bar{L} \right)\end{aligned}$$

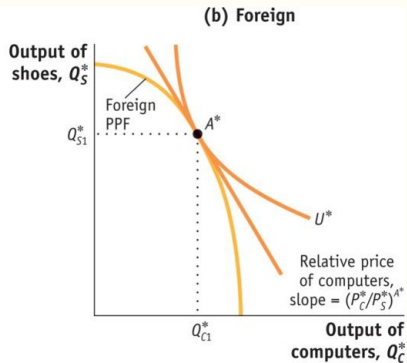
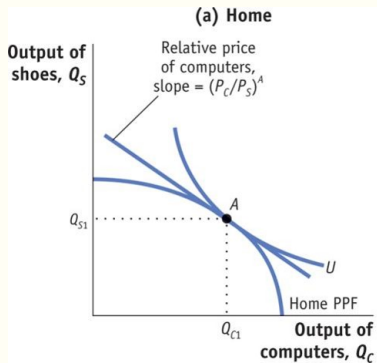
- What if the economy is endowed with more capital $\bar{K}$? More labor $\bar{L}$?
- What if the relative price of Computers $\frac{P_C}{P_S}$ increases? Decreases?

Relative Supply

- As can be seen above, *generally, an economy will tend to be relatively effective at producing goods that are intensive in the factors with which the country is relatively well endowed.*
- This can be seen explicitly by solving for the relative supply function

$$\frac{Q_C}{Q_S} = \frac{\left(3 \left(\frac{\bar{K}}{\bar{L}} \right) \left(\frac{P_C}{P_S} \right)^2 - 1 \right)}{3^{\frac{3}{4}} \left(3 - \left(\frac{\bar{K}}{\bar{L}} \right) \left(\frac{P_C}{P_S} \right)^2 \right) \left(\frac{P_C}{P_S} \right)}$$

- What if $\frac{\bar{K}}{\bar{L}} > \frac{\bar{K}^*}{\bar{L}^*}$?
- But, perhaps more intuition can come from the PPFs



Home Relative Supply

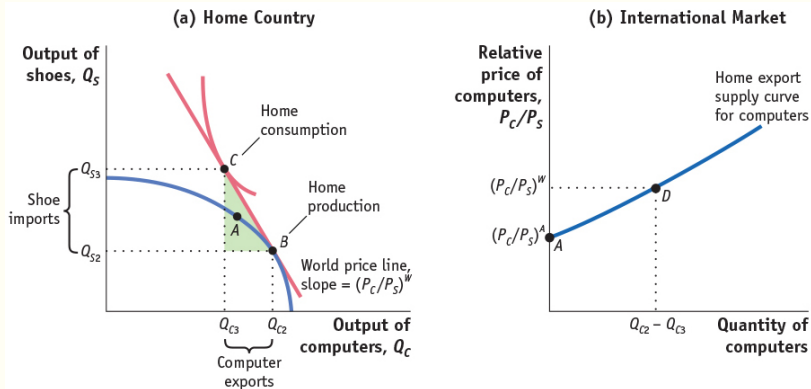


Figure 4.3

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Foreign Relative Demand

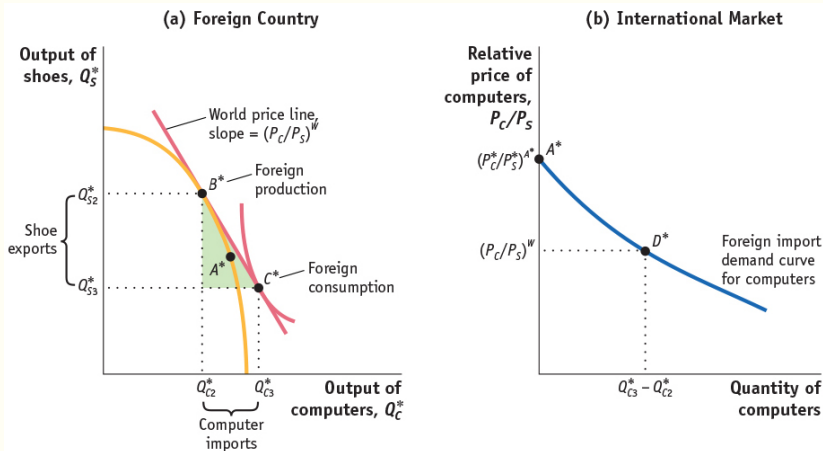


Figure 4.4

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Trade Equilibrium

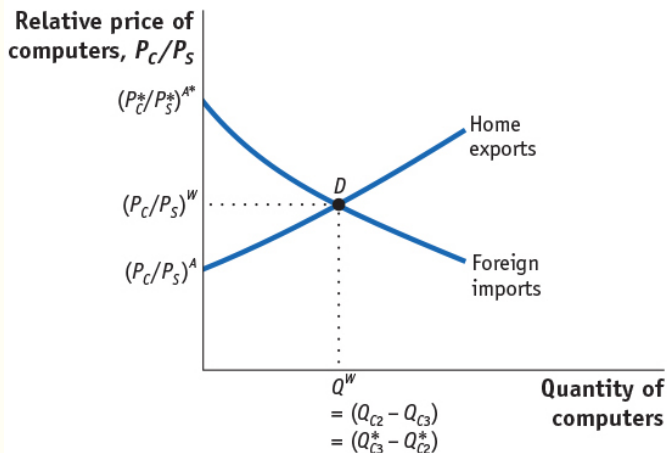


Figure 4.5

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- **The H-O Theorem:** A country will have comparative advantage in, and therefore will export, that good whose production is relatively intensive in the factor with which that country is relatively well endowed.
- **The Rybczynski Theorem:** At constant world prices, if a country experiences an increase in the supply of one factor, it will produce more of the product whose production is intensive in that factor and less of the other.

- **The Factor Price Equalization Theorem:** Given all the assumptions of the H-O model, free international trade will lead to the international equalization of individual factor prices.

→ 3 reasons why this might fail...

- ❶ If countries have different technologies of production.
- ❷ Needs complete convergence of the prices of goods. In the real world, prices of goods are not fully equalized by international trade – both natural barriers (such as transportation costs) and government (such as tariffs, import quotas, and other restrictions).
- ❸ We also need the countries to produce the same set of goods. One possible reason this isn't the case is countries have radically different ratios of capital to labor or of skilled to unskilled labor.

H-O Model Assumptions Recap

ASSUMPTIONS:

- 1 Labor and capital flow freely between the industries.
- 2 The production of shoes is labor-intensive as compared with computer production, which is capital-intensive.
- 3 The amounts of labor and capital found in the two countries differ, with Foreign abundant in labor and Home abundant in capital.
- 4 There is free international trade in goods.
- 5 The technologies for producing shoes and computers are the same across countries.
- 6 Tastes are the same across countries.

Testing the Heckscher-Ohlin Model

The first test of the Heckscher-Ohlin theorem was performed by economist Wassily Leontief in 1953.

- Leontief supposed correctly that in 1947 the United States was abundant in capital relative to the rest of the world.
 - Leontief assumed that U.S. and foreign technologies were the same due to the limited data on foreign technology, which is consistent with HO theorem.
- Thus, from the Heckscher-Ohlin theorem, Leontief expected that the United States would export capital-intensive goods and import labor-intensive goods.
- What Leontief actually found, however, was just the opposite: the capital-labor ratio for U.S. imports was higher than the capital-labor ratio found for U.S. exports.
- This finding contradicted the Heckscher-Ohlin theorem and came to be called **Leontief's paradox**.

Leontief's (1953) Test

	Exports	Imports
Capital (\$ million)	\$2.5	\$3.1
Labor (person-years)	182	170
Capital/labor (\$/person)	\$13,700	\$18,200

- U.S. and foreign technologies are not the same.
- By focusing only on labor and capital, Leontief ignored land.
- Labor should have been disaggregated by skill (since it would not be surprising to find that U.S. exports are intensive in skilled labor).
- The data for 1947 may be unusual, since World War II had just ended.
- The U.S. was not engaged in free trade, as the HO model assumes.

Understanding Leontief's Paradox

- As with any result that seems counterintuitive, you should always ask yourself a couple questions.
- Is the methodology correct? In particular, if the methodology is appealing to a theoretical model, were the crimes committed in moving from theory to data significant?
- If the methodology is correct, are there alternative explanations outside the model which might explain the result?
- If exploring both questions leads you to accept the methodology and the result, then you have a real puzzle
- As you will notice, there is a common theme in economics that “puzzles” are really just bad methodology or bad models (smell test usually sufficient to identify a faux puzzle)

Testing the Heckscher-Ohlin Model

- An important study by Donald Davis and David Weinstein showed that if relax the assumption of common technologies, along with assumptions underlying factor price equalization (countries produce the same goods and costless trade equalizes prices of goods):
 - then the predictions for the direction and volume of the factor content of trade line-up well with empirical evidence and ultimately generate a good fit.
- Difficulty finding support for the predictions of the “pure” Heckscher-Ohlin model can be blamed on some of the assumptions made.

Davis and Weinstein (2001)

	Assumptions Dropped*			
	None	Drop (1)	Drop (1)–(2)	Drop (1)–(3)
Predictive Success (sign test)	0.32	0.50	0.86	0.91
Missing Trade (observed/ predicted)	0.0005	0.008	0.19	0.69

*Assumptions: (1) common technologies across countries; (2) countries produce the same set of goods; and (3) costless trade equalizes goods prices.

Source: Don R. Davis and David Weinstein, "An Account of Global Factor Trade," *American Economic Review* (2001), pp. 1423–1453.

Effect of Trade on Factor Prices

- Let's solve for equilibrium wages and rental rates and for simplicity focus just on the Computer industry (note wages and rental rates have to be the same across industries)

$$w = P_C MPL_C$$

$$r = P_C MPK_C$$

- Continue to use our example from before

$$Q_S = L_S^{\frac{3}{4}} K_S^{\frac{1}{4}}$$

$$Q_C = L_C^{\frac{1}{4}} K_C^{\frac{3}{4}}$$

Effect of Trade on Factor Prices

- Recall that in equilibrium, it must be the case that

$$\begin{aligned}\frac{P_C}{P_S} &= \frac{MPL_S}{MPL_C} = 3 \left(\frac{K_S}{L_S} \right)^{\frac{1}{4}} \left(\frac{L_C}{K_C} \right)^{\frac{3}{4}} \\ &= 3 \left(\frac{K_C}{9L_C} \right)^{\frac{1}{4}} \left(\frac{L_C}{K_C} \right)^{\frac{3}{4}} \\ &= \left(\frac{3L_C}{K_C} \right)^{\frac{1}{2}}\end{aligned}$$

- Which means that

$$\frac{L_C}{K_C} = \frac{1}{3} \left(\frac{P_C}{P_S} \right)^2 \quad \text{or} \quad \frac{K_C}{L_C} = 3 \left(\frac{P_S}{P_C} \right)^2$$

Effect of Trade on Factor Prices

- Recall our factor price equations (and convert to real prices)

$$\frac{w}{P_C} = MPL_C \quad \text{and} \quad \frac{r}{P_C} = MPK_C$$

$$\Rightarrow \frac{w}{P_C} = \frac{1}{4} \left(\frac{K_C}{L_C} \right)^{\frac{3}{4}} \quad \Rightarrow \frac{r}{P_C} = \frac{3}{4} \left(\frac{L_C}{K_C} \right)^{\frac{1}{4}}$$

$$\Rightarrow \frac{w}{P_C} = \frac{1}{4} \left(3 \left(\frac{P_S}{P_C} \right)^2 \right)^{\frac{3}{4}} \quad \Rightarrow \frac{r}{P_C} = \frac{3}{4} \left(\frac{1}{3} \left(\frac{P_C}{P_S} \right)^2 \right)^{\frac{1}{4}}$$

$$\therefore \frac{w}{P_C} = \frac{3^{\frac{3}{4}}}{4} \left(\frac{1}{\frac{P_C}{P_S}} \right)^{\frac{3}{2}} \quad \therefore \frac{r}{P_C} = \frac{3^{\frac{3}{4}}}{4} \left(\frac{P_C}{P_S} \right)^{\frac{1}{2}}$$

Effect of Trade on Factor Prices

$$\frac{w}{P_C} = \frac{3^{\frac{3}{4}}}{4} \left(\frac{1}{\frac{P_C}{P_S}} \right)^{\frac{3}{2}}$$

$$\frac{r}{P_C} = \frac{3^{\frac{3}{4}}}{4} \left(\frac{P_C}{P_S} \right)^{\frac{1}{2}}$$

- What happens if the home country – which is capital abundant – opens up to trade?
- If the home country does trade, what do we know has to be true about the relative price of Computers $\frac{P_C}{P_S}$?
- $\frac{P_C}{P_S} \uparrow \Rightarrow \frac{w}{P_C} \downarrow$ and $\frac{r}{P_C} \uparrow$.
- **The Stolper-Samuelson Theorem:** Free international trade benefits the abundant factor and harms the scarce factor.

Opinions Toward Free Trade

- In the short-run specific-factors model, labor may gain or lose from free trade, but the specific factor in the export sector gains, and the specific factor in the import sector loses.
- If labor also earns some part of the rental on the specific factor in their industry, the *industry of employment* of workers will affect their attitudes toward free trade.
- In the long-run Heckscher-Ohlin model, however, the industry of employment should not matter.
- According to the Stolper-Samuelson theorem, an increase in the relative price of exports will benefit the factor of production used intensively in exports and harm the other, regardless of the industry in which these factors of production actually work.

Opinions Toward Free Trade

- In the United States, export industries tend to use high-skilled labor intensively for research and development and other scientific work.
- Therefore, an increase in the relative price of exports will benefit skilled labor in the long run, regardless of whether these workers are employed in export-oriented industries or import-competing industries.
- In the long run, then, the skill level of workers should determine their attitudes toward free trade.

Conclusion

- We have now seen 4 standard theoretical trade models
- Adam Smith → Ricardian → Specific-Factors → Heckscher-Ohlin
- S.F. and H.O. models shed light on how some may lose from freer trade despite the aggregate winning.
- Next up, we dive a bit deeper into who loses from trade.
- Keep in mind that though there are losers to freer trade, trade protection is typically seen as an inefficient mechanism to help those who lose; i.e.
 - (1) Hufbauer and Lowry (2012) finds that in the case of tariffs set on tires by Obama in 2009, “the cost per job manufacturing saved (a maximum of 1,200 jobs by our calculations) was at least \$900,000 in that year.”
 - (2) Amiti et. al (2019) finds that the 2017-2018 trade war has resulted in (conservatively) \$200,000 welfare loss per job saved.